



Figure 5: Skifta app displaying media servers

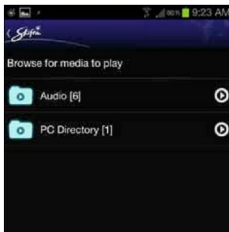


Figure 6: Files shared using MediaTomb



Figure 7: DLNA.org Android app

Figure 3, you can see MediaTomb (running on a Linux box), a NAS server (*axentraserver:root*), Rygel (another media server for Linux), an Android mobile (SAMSUNG GT-N7000), a Belkin router (with UPnP support) and experimental GUPnP Network Light. This application also gives you details about the interaction between the devices, the state changes and services offered. The GUPnP Network Light sample application simulates a UPnP device on the network, letting you toggle between the two possible states, ON and OFF.

UPnP Router Control is an application that connects to the home media gateway (router) that all devices are connected through; it displays upload/download speeds, real-time network activity and the total volume of media data exchange (Figure 4).

Android apps

Your Android phone can play media stored on a DMS (like a PC or NAS storage device). Skifta is one DLNA player app; it automatically detects DMSs on the network and uses the Android default player or user-installed players for rendering. Other popular Android DMS/DMP apps are the aVIA media player, the MediaHouse UPnP/DLNA browser and bubbleUPnP, which let you share media from your mobile with other devices.

Scenarios

What if the TV you currently have is not DLNA-certified, or if it does not have an Ethernet port or Wi-Fi? Can you not experience DLNA features? There is a solution, through additional devices like iOmega TV Link, or media players that are DLNA-capable and can be connected to the TV using the AV cable or HDMI. There are also products that combine storage and media players with A/V output and HDMI—one of these is the iOmega ScreenPlay DX HD Media player. Another option is to use gaming consoles like Xbox and Sony PlayStation 3, which are DLNA certified.

To share a huge media collection with your PC, you have

to keep your PC on all the time; the alternative is to buy a product like the Seagate GoFlex Home Network Storage system, instead of USB external disks.

Check DLNA certification online

If you did not pay attention to the DLNA / UPnP logos on the product package, you can also check online for the full specifications of the product, or you can search the DLNA home page (*dlna.org*) for certification.

New buys

If you are buying a new TV, Blu-ray player or a mobile, be sure it has DLNA certification, or uses a platform that has DLNA software available. When you buy home networking devices like Wi-Fi routers or NAS storage devices, check the specifications for DLNA or UPnP. Figure 7 shows an Android app from DLNA.org that will help you make a choice if you are planning to buy new devices.

When you are buying a Wi-Fi router, you might find lower-speed routers (e.g., 54 MBps) cheaper. After all, there's a long wait for Net connection speeds to reach that level, affordably. But it might be better to have a higher-speed router (like 150 MBps+) for your home networking needs, as it would make video streaming to your HD TV smooth. **END** 

References

- [1] DLNA technology tutorial: http://www.sony.net/SonyInfo/technology/technology/theme/dlna_01.html
- [2] Open source GUPnP project: <https://live.gnome.org/GUPnP/>
- [3] DLNA product certification search: <http://www.dlna.org/dlna-for-industry/digital-living/product-search>

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